gheim: A Swiss-Tuned PII Detector and Companion Dataset for the Four Official Languages of Switzerland and English

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We release gheim-ch-pii-171k, a 171,336-chunk personally-identifiable information (PII) named-entity-recognition dataset covering the four official Swiss languages (Swiss German, Swiss French, Swiss Italian, Romansh) and English, and gheim-ch-560m, a 560M-parameter xlm-roberta-large fine-tune released alongside it. The dataset combines real Swiss text from court rulings, federal parliament records, Swiss-filtered web text and a Romansh corpus with an LLM-generated synthetic gap-fill layer that covers cells where the real-text corpus was too sparse for training and evaluation. The model attains a strict-span F_1 of 0.916 on a held-out test split, outperforming the next-strongest open contestant (openai/privacy-filter zero-shot) by 47 percentage points on the same Swiss test set, and reaching 0.89–0.96 person-NER F_1 on four widely-used external benchmarks. The dataset ships under CC BY 4.0; the model under Apache 2.0; both are available on the Hugging Face Hub. Reproducible eval scripts and checkpoints are on GitHub.

1 INTRODUCTION

The dominant pattern in production large-language-model (LLM) workflows is to call a hosted API: OpenAI, Anthropic, Google, Mistral. For Swiss-market deployments this raises a recurring compliance concern: every request that contains a customer name, a Swiss address, an IBAN, or an AHV/AVS social-security number leaves Swiss jurisdiction the moment it crosses the API boundary. Anonymising sensitive content before the request and restoring the originals on the response is the established mitigation. The accuracy of the round-trip depends on the quality of the underlying named-entity recogniser.

Existing public PII detectors handle Swiss text poorly. The closest open generalist, openai/privacy-filter [11], reaches a strict-span F_1 of 0.44 on our held-out Swiss test set despite being a 1.4B-parameter mixture-of-experts model trained on multilingual data. Microsoft Presidio [9], configured with per-language NLP engines, reaches 0.43. Multilingual NER baselines such as Davlan's xlm-roberta-base-ner-hrl [1] reach 0.65 person-only F_1. None of the available systems fluently handle the four official Swiss languages together with the structured PII formats common in Swiss documents (CH-IBAN, AHV, VAT-CHE), and only one (SwissNER [17]) provides a Romansh evaluation slice.

This work contributes:

(1) gheim-ch-pii-171k, a multilingual PII NER dataset of 171,336 chunks across Swiss German, Swiss French, Swiss Italian, Romansh and English, covering eight PII categories (account number, address, date, email, person, phone, URL, secret), encoded in a 33-class BIOES tag scheme byte-aligned with openai/privacy-filter.
(2) gheim-ch-560m, a 560M-parameter token-classification model fine-tuned from xlm-roberta-large [4] on the dataset, attaining a held-out test F_1 of 0.916 strict-span and 0.958 character-level.
(3) A reproducible evaluation harness that scores both the released model on four established external benchmarks (ai4privacy/openpii-1m, ZurichNLP swissner, Babelscape WikiNeural [14], CoNLL-2003 [15]) and seven competing open detectors on the released test set, with replication checks against each baseline's published numbers on its native dataset.

The rest of the paper is structured as follows. Section 2 describes the dataset construction. Section 3 covers model selection and training. Section 4 reports the evaluation, including the methodology-validation cross-checks. Section 5 discusses limitations. Section 6 situates the work against prior art. Section 7 concludes.

2 DATASET: GHEIM-CH-PII-171K

2.1 Source corpora and language coverage

Real-text content is drawn from the Apertus pretrain corpora [13], specifically the Swiss legal slice (`entscheidsuche`), the federal parliament records (`curia_vista`), the Swiss-filtered subset of FineWeb-2, and the Romansh corpus (`romansh`). Source documents are chunked at sentence boundaries to roughly 1,500 characters, then tagged with `facebook/fasttext-language-identification` at confidence ≥ 0.5. Chunks below threshold are dropped. The five-language distribution of the released dataset is shown in Table 1.

Table 1. Per-language chunk distribution.

Code	Variant	Chunks	%
de_ch	Swiss German written standard	48,564	28.3
fr_ch	Swiss French	48,097	28.1
it_ch	Swiss Italian (Ticino/Grisons)	45,462	26.5
rm	Romansh (Rumantsch Grischun + idioms)	17,171	10.0
en	English (international, Swiss-context)	12,042	7.0
Total		171,336	100

Spoken-form Swiss German (GSW) is not represented: the corpus is written-standard German as used in Swiss legal and journalistic contexts. The fasttext language identifier labels GSW ("chuchichäschtli"-style orthography) as standard German, so incidental dialect content cannot be reliably separated.

2.2 Annotation schema

The label space follows the openai/privacy-filter taxonomy: eight PII categories listed in Table 2, each encoded in a BIOES tag scheme [12] (Begin, Inside, Outside, End, Single), giving 33 classes overall (one outside class plus four BIOES positions for each of the eight categories). The byte-identical label mapping is deliberate: it lets a fine-tune started from openai/privacy-filter's checkpoint reuse the pretrained classifier-head weights without remapping. The `private_` prefix is historical; all eight cells are applied with a recall-oriented labelling policy and may flag publicly-listed institutional entities (e.g. a court switchboard phone number).

Table 2. The eight PII categories.

Category	Examples
`account_number`	CH-IBAN, AHV/AVS, VAT-CHE, credit cards
`private_address`	Bahnhofstrasse 1, 8001 Zürich
`private_date`	1990-01-02, 14. März 2026
`private_email`	alice@example.ch
`private_person`	Joel Barmettler, Dr. Schmidt
`private_phone`	+41 44 268 12 34
`private_url`	https://kanzlei.ch/...
`secret`	API keys, bearer tokens

2.3 Curation pipeline

The dataset is the output of a seven-phase pipeline. Real-text chunks are passed through (i) language detection, (ii) primary annotation by Gemma 4 26B-A4B via structured-output JSON-schema decoding, (iii) augmentation by a regex catalogue with checksum validators (ISO 13616 mod-97 for IBAN, EAN-13 mod-10 for AHV, ISO 7064 mod-11 for VAT-CHE, Luhn for credit cards), with the regex annotator taking priority on overlap. The annotated chunks are then balanced by per-document, per-value and per-cell caps to limit document-level concentration and value memorisation, then split at the document level into train, validation and test stratified by (language $\times$ source).

A separate synthetic gap-fill layer is appended to populate cells where real-Swiss-text coverage was insufficient. A slot-fill verifier pre-generates payload values (Faker_CH names, addresses and phones; checksum-validated CH-IBAN, AHV and VAT-CHE; common secret formats), then prompts Gemma to embed those exact values verbatim into natural multilingual prose (developer chat, customer-record notes, incident reports). The verifier accepts a chunk only if every payload value appears as an exact substring in the model output. Rejection rate was approximately 2 % of generations. 24,900 of these synthetic chunks are placed in the training split (22,500), validation (1,200) and test (1,200), with deterministic seeded placement and disjoint chunk IDs across splits. The remaining 146,436 chunks are real-text. Every record carries a `synthetic` boolean field so consumers can filter to real-only or synthetic-only as required.

2.4 Splits and held-out integrity

Documents are partitioned at the source-document level before any chunk is taken, with stratification by (language × source), so no chunk in test or validation comes from a document with a chunk in train. Synthetic chunks are placed deterministically (random seed 17). Final split sizes are 139,641 / 15,834 / 15,861 chunks for train / validation / test.

2.5 Label noise

Annotations are machine-generated; no third-party human annotators were contracted. As an upper bound on label noise we ran four independent subagent re-labellings on a stratified 176-chunk sample and compared the dataset to the four-way majority vote, yielding an F_1 of 0.66. Inspection of disagreements indicates the gap is dominated by policy-interpretation differences (e.g. whether to label the publicly-listed switchboard number of a court as PII) rather than random error.

3 MODEL: GHEIM-CH-560M

3.1 Architecture and base-model selection

The model is a token-classification head atop xlm-roberta-large [4] (24 layers, 1024 hidden, 16 heads, 560M parameters). XLM-R-large was selected following a controlled bake-off against ZurichNLP/swissbert [(270M dense), summarised in Table 3. Both bases received an identical 5×3 hyperparameter sweep over (learning rate, layer-wise LR decay) at one epoch, after which the winning configuration per base was trained for three full epochs and selected by best validation F_1. A third candidate, openai/privacy-filter, was deferred: the Olmo-MoE ONNX export is not currently stable in optimum [8], which would constrain JavaScript-side portability of the released model.

Table 3. Base-model bake-off.

Base model	Val F_1	Test F_1
ZurichNLP/swissbert (270M)	0.910	not run
FacebookAI/xlm-roberta-large (560M)	**0.918**	**0.916**

In the (LR, LLRD) sweep, learning rate was the dominant factor: F_1 increased monotonically with LR over the tested range (5e-6 to 5e-5) for both bases. Layer-wise LR decay did not improve any cell at any tested value (1.0, 0.95, 0.9). The selected configuration is therefore LR = 5×10^{-5} without LLRD.

3.2 Training procedure

The training mix is the train split of gheim-ch-pii-171k (139,641 chunks) plus a small AI4Privacy English/email augmentation slice (approximately 14,000 chunks drawn from `ai4privacy/openpii-1m` [3]) added to shore up English-anchor coverage and CH-region email recall, for a total of 153,641 training chunks. The validation and test splits used for the headline metrics contain no AI4Privacy data. Hyperparameters are listed in Table 4.

Table 4. Training hyperparameters.

Hyperparameter	Value
Optimiser	AdamW [7]
Learning rate	5e-5
LR scheduler	Cosine, 5% warmup
Layer-wise LR decay	1.0 (off)
Effective batch size	128 (per-device 64 × 2 GPUs DDP)
Epochs	3; best at epoch 2.50
Max sequence length	512
Precision	bf16
Hardware	2 × NVIDIA RTX 4090
Wall time	≈ 52 min train + 4 min eval

The best checkpoint was selected by `eval_overall_f1` on the validation split. The held-out test split was scored exactly once on the selected checkpoint to produce the headline numbers in Section 4.

3.3 Deployment formats

The model ships in two formats. Server-side users load `model.safetensors` (fp32 PyTorch, 2.2 GB) via transformers [18]; in-browser users load `onnx/model_quantized.onnx` (int8 dynamic-quantised ONNX, 552 MB) via transformers.js [6] on WebGPU or WebAssembly. The int8 export reproduces 99.4% of the fp32 strict F_1 (0.909 vs 0.916, −0.7 pp); the loss is concentrated in the two categories that were already weakest under fp32 (Table 5, columns headed "Quant. Δ").

4 EVALUATION

4.1 Methodology

Two metrics are reported throughout. **Strict-span $\mathbf{F_1}$** is the standard NER literature metric: a predicted span counts as a true positive only if its (start, end, label) tuple matches a gold span exactly. Computed via seqeval [10] at the token level for HF and ONNX backends, and via char-level set match for span-emitting backends (spaCy, Presidio) that lack a token grid. **Char-level $\mathbf{F_1}$** (label-aware) is computed over (character index, label) pairs: it asks "did the model mask the right characters with the right category?" and is invariant to entity-segmentation policy differences across systems.

The latter matters because cross-model PII evaluation has a structural problem: different annotation conventions split the same surface form differently. ai4privacy/openpii-1m splits a person mention into a `GIVENNAME` and a `SURNAME` span; gheim-ch-560m and CoNLL-2003 emit one combined `PERSON` span; address detectors fragment "Werdstrasse 36, 8004 Zürich" at the comma or do not. Strict-span F_1 penalises every boundary mismatch even when the underlying redaction-utility goal is met. Reporting both metrics lets the reader judge which is appropriate for their use case: strict for literature comparability, char for deployment.

4.2 Headline performance

On the held-out test split of gheim-ch-pii-171k (15,861 chunks), gheim-ch-560m attains a strict-span F_1 of 0.916 (precision 0.907, recall 0.926) and a char F_1 of 0.958. The validation/test gap is 0.002, consistent with no observable overfitting to the validation set used for checkpoint selection. Per-category and per-language breakdowns are in Tables 5 and 6.

Table 5. Per-category F_1. "Quant. Δ" is the change under int8 quantisation relative to the fp32 strict cell.

Category	Strict	Char	Quant. Δ	n_{gold}
`secret`	0.996	0.998	+0.002	1,104
`private_email`	0.987	0.994	−0.003	1,712
`private_phone`	0.986	0.989	−0.002	3,329
`private_url`	0.947	0.991	−0.002	3,991
`private_date`	0.922	0.935	−0.007	9,678
`private_person`	0.915	0.944	−0.006	14,726
`private_address`	0.782	0.886	−0.021	3,507
`account_number`	0.669	0.763	−0.053	502

The best-served categories are the structured ones with strong surface signals (`secret`, `email`, `phone`, `URL`); the weakest are `private_address` and `account_number`. Address weakness is dominated by boundary placement on multi-token addresses; account-number weakness reflects the

Table 6. Per-language F_1 on the held-out test split.

Language	Strict	Char	n_{chunks}
en	0.954	0.981	893
it_ch	0.942	0.972	4,169
de_ch	0.911	0.954	4,753
fr_ch	0.906	0.953	4,705
rm	0.853	0.923	1,341

Table 7. Direction A: other models on the gheim-ch-pii-171k test split. Strict and char F_1 for both the overall 8-category metric and the `private_person` cell.

Model	Strict F1	Char F1	PER strict	PER char
joelbarmettler/gheim-ch-560m	**0.916**	**0.958**	**0.915**	**0.944**
openai/privacy-filter (1.4B MoE, zero-shot)	0.443	0.610	0.406	0.561
Microsoft Presidio (multi-lang config)	0.434	0.562	0.442	0.521
Davlan/xlm-roberta-base-ner-hrl	n/a	n/a	0.645	0.728
Davlan/distilbert-base-multilingual-cased-ner-hrl	n/a	n/a	0.619	0.771
spaCy `de/fr/it_core_news_lg`	n/a	n/a	0.505	0.621
dslim/bert-base-NER (English slice)	n/a	n/a	0.512	0.770
Isotonic/distilbert_finetuned_ai4privacy_v2	0.157	0.508	0.264	0.507

small per-language support (502 spans across all five languages) plus the structural similarity of CH-IBAN, EU-IBAN and other non-Swiss numbers. For production use both categories are intended to be paired with a deterministic regex front-end that applies checksum validation (see `packages/gheim-py/src/gheim/detectors/composite.py`).

4.3 Comparison to other systems

Seven other open PII / NER systems were scored on the same test split under the same harness. Results are in Table 7. PER-only models (Davlan, dslim, spaCy) emit only person mentions and no other PII categories; their overall F_1 on our 8-category schema is therefore artificially low and is reported as n/a in the table; the meaningful number for those is the `private_person` cell (Table 7, last two columns).

The released model leads every other system on every cell that is directly comparable. The two metrics tell the same story: strict F_1 shows a 47-pp lead over the strongest contestant (openai/privacy-filter), char F_1 shows a 35-pp lead. The gap between the two metrics is largest for systems whose schema differs most from ours: Isotonic/distilbert_finetuned_ai4privacy_v2 gains +35 pp under char F_1 because its fine-grained schema (FIRSTNAME, LASTNAME, MIDDLENAME, PREFIX, . . .)

collapses under our 8-category remap and produces highly fragmented spans; openai/privacy-filter gains +17 pp because the Olmo tokeniser emits subword boundaries that do not align with the gold span ends.

Per-language overall char F_1 is given in Table 8: gheim leads every other system in every language. The model is the only entry with native Romansh support.

Table 8. Per-language char F_1 on the gheim-ch-pii-171k test split, top three contestants.

Lang.	gheim	priv-filter	Presidio
en	**0.981**	0.788	0.561
de_ch	**0.954**	0.569	0.572
fr_ch	**0.953**	0.577	0.596
it_ch	**0.972**	0.630	0.690
rm	**0.923**	0.643	0.251

4.4 Cross-domain evaluation

The released model was scored on four established external benchmarks (Table 9). For pure-NER datasets (swissner, CoNLL-2003, WikiNeural) only PER maps to a gheim category; the meaningful number is therefore the PER cell. The non-PER overall F_1 is dragged down by the model predicting categories (dates, URLs, addresses) that the NER datasets do not label.

The model's PER F_1 stays in [0.89, 0.96] across all four datasets under both metrics. This is consistent with no overfitting to the training distribution: the four datasets cover templated synthetic English+European PII (openpii-1m), real Swiss news in four languages (swissner), the 22-year-old reference English NER benchmark (CoNLL-2003), and Wikipedia-grounded multilingual NER (WikiNeural). The notable exception is the strict-span PER score on openpii-1m (0.776 vs. char 0.920): openpii-1m's gold splits each person mention into separate `GIVENNAME` and `SURNAME` spans, which after our remap fragment into two adjacent spans of `private_person` where our model emits one combined span. Char F_1 reports the redaction outcome (every PII character correctly masked) and is the appropriate metric for the cross-schema comparison.

On Swiss news (swissner, the only Swiss-news NER set that includes Romansh), per-language person-character F_1 is 0.864 (de), 0.870 (fr), 0.835 (it), 0.795 (rm).

4.5 Methodology validation

To check that the harness above is sound rather than an unfair implementation, we replicated each external system on the dataset on which it was trained and compared our reproduction to the system's own published F_1 (Table 10). Three of four reproductions match within ±0.03; the fourth (Isotonic) is discussed below.

Table 9. Direction B: gheim-ch-560m on external benchmarks.

External benchmark	n	Overall strict	Overall char	PER strict	PER char
ai4privacy/pii-masking-openpii-1m	8,000	0.919	0.972	0.776	0.920
Babelscape WikiNeural	8,000	0.833	0.880	0.919	0.960
ZurichNLP swissner	800	0.781	0.838	0.903	0.946
CoNLL-2003	3,453	0.678	0.703	0.887	0.936

Table 10. Methodology validation: each baseline scored on its own training-distribution dataset, against published numbers.

Replication target	Their claim	Our repro	Δ
dslim/bert-base-NER on CoNLL-2003	0.910	0.913	+0.003
dslim PER on CoNLL-2003	~0.96	0.957	−0.003
Davlan-XLMR on WikiNeural avg	~0.84	0.813	−0.027
spaCy `de_core_news_lg` on swissner_de	~0.83	0.841	+0.011
Isotonic distilbert on AI4Privacy	0.955	0.969[†]	+0.014

[†]The fourth row required investigation. Isotonic's claimed F_1 of 0.955 initially reproduced as 0.78 strict-span F_1 on the same data, an apparent 17-pp gap. The difference traces to the metric: Isotonic's reported number is *per-token classification accuracy* (which includes the dominant `O` class and is much higher than per-span F_1). Reproducing Isotonic's metric exactly (per-token accuracy including `O`) yields 0.969, matching their 0.955 within sampling noise. Our strict-span F_1 of 0.78 on the same data is the correct NER-literature comparison.

5 LIMITATIONS

- **Recall-oriented labelling policy.** Public Swiss documents contain extensive personally-identifiable information that is already public (court rulings, parliament records, public officials). The dataset labels these as PII and the model learns to flag them. Applications that need to distinguish public from private entities should layer a downstream public-vs-private classifier.
- **`private_address` test F_1 is 0.78.** Boundary placement on multi-token addresses is the dominant error mode.
- **`account_number` test F_1 is 0.67.** Production deployments should pair the model with a regex front-end with checksum validation (IBAN, AHV, VAT-CHE, Luhn).
- **Romansh test F_1 is 0.85**, the weakest of the five languages. The Romansh training material is dominated by a single literary/journalistic register (Rumantsch Grischun), so dialectal or technical RM text is unmeasured.

- **Swiss German dialect (GSW) is not measured.** The fasttext detector used in data preparation labels GSW as standard German.
- **Annotations are machine-generated.** The dataset was not human-annotated; the F_1 of 0.66 against a four-way subagent re-labelling is reported as an upper bound on label noise rather than as a quality claim.
- **Re-identification is not in scope.** The model is intended for redaction. It is not designed to identify or link entities to real persons and does not return entity-linked identifiers.

6 RELATED WORK

General-purpose PII detectors. Microsoft Presidio [9] is the dominant rule-and-NER hybrid in industry: a per-language spaCy NER engine combined with regex and checksum recognisers for structured formats. The AI4Privacy family [2, 3] provides large-scale templated synthetic PII corpora and a number of fine-tuned distilbert checkpoints; their breadth (12+ extended categories, 17+ locales) is greater than ours but real-text Swiss coverage is absent. openai/privacy-filter [11] is a 1.4B mixture-of-experts checkpoint (50M active parameters) that shares our 33-class BIOES schema and is the canonical zero-shot baseline for our comparison.

Multilingual NER.. The XLM-R family [4] is the de-facto starting point for multilingual token-classification. Davlan's high-resource-language NER fine-tunes [1] are trained on PER/ORG/LOC across many European languages and serve as a generalist baseline. WikiNeural [14] provides large-scale silver-data multilingual NER training and evaluation. The classic CoNLL-2003 shared task [15] remains the reference English NER benchmark.

Swiss NLP.. SwissBERT [16] is a multilingual encoder pretrained on Swiss text covering all four official languages, and its companion SwissNER [17] is the only public Swiss news NER test set with a Romansh slice. The Apertus pretrain corpora [13] provide the source text for our real-chunk content. Faker_CH provides Swiss-locale random data for our synthetic gap-fill payloads.

Industry tooling. transformers [18] is the runtime; transformers.js [6] powers the browser-side demo; seqeval [10] provides the strict-span scoring; spaCy [5] provides the per-language NER pipelines used in the comparison and inside Presidio.

7 CONCLUSION

We released gheim-ch-pii-171k and gheim-ch-560m, a Swiss-tuned PII NER dataset and model targeting a deployment context (LLM-API anonymisation round-trip) that the existing open detectors handle poorly on Swiss text. The model leads every public contestant by 35-pp on the same Swiss test set under either reported metric and stays within $[0.89, 0.96]$ person-NER F_1

across four cross-domain external benchmarks. The companion dataset is released under CC BY 4.0; the model under Apache 2.0; both are on the Hugging Face Hub. The full evaluation harness, the per-system JSONs, and the reproducible build of the comparison tables are at https://github.com/joelbarmettlerUZH/gheim.

Future work. A privately-tuned variant covering spoken GSW, dialectal Romansh and additional account-number formats; a distilled student model in the 100M-parameter range for edge-side deployment; and a public-versus-private-entity post-classifier to relax the recall-oriented labelling policy when applications require stricter precision.

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